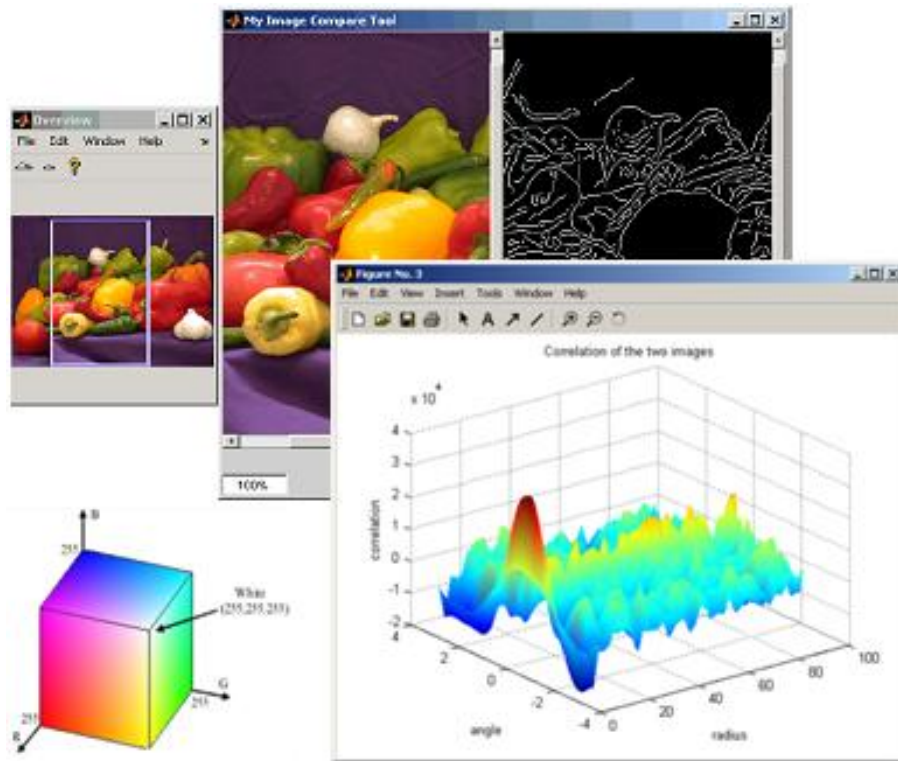


Digital Image Processing

LECTURE 13: MORPHOLOGICAL OPERATION



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Morphological Operations

Morphological Operations

- ❑ The word ***morphology*** commonly denotes a branch of biology that deals with the form and structure of animals and plants.

Morphological Operations

- ❑ Image Morphology is the study of shapes of the objects present in the image and extraction of image features. Image features are necessary for object recognition.
- ❑ We use the same word here in the context of mathematical morphology as a tool for **extracting image components** that are useful in the **representation and description of region shape**, such as **boundaries, skeletons** and **convex hull**.

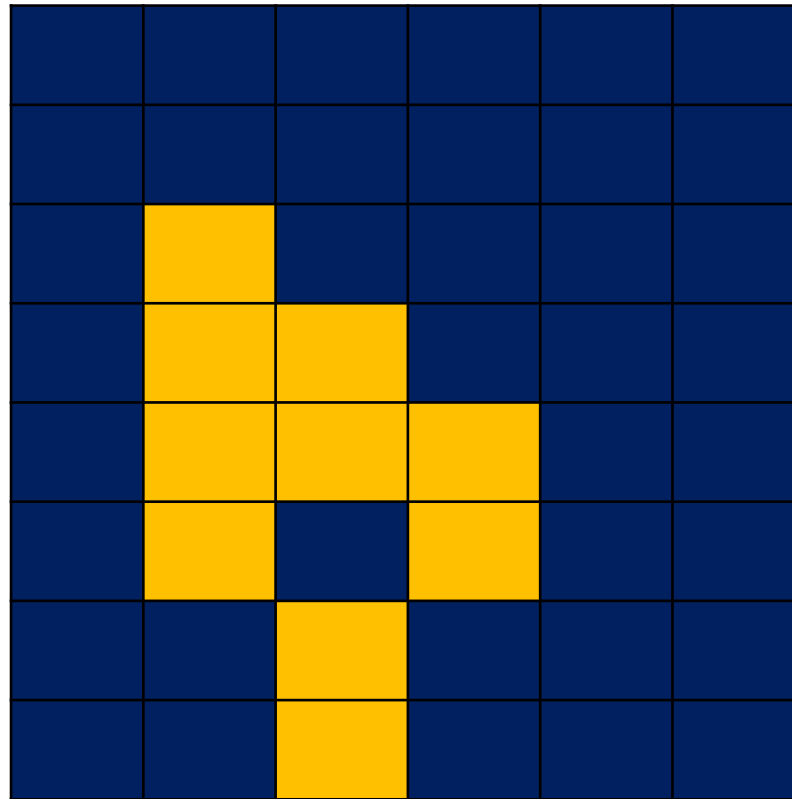
Morphological Operations

□ Applications of Morphological Operations:

- Enhancing object structure
 - Skeletonization, thinning, thickening, convex hull and object marking etc.
- Object quantifications
 - Determination of Area, Perimeter and boundary of an image
- Pre-processing
 - Filtering
 - Shape simplification
- Segmentation using object shape

Morphological Operations

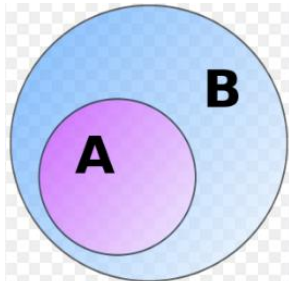
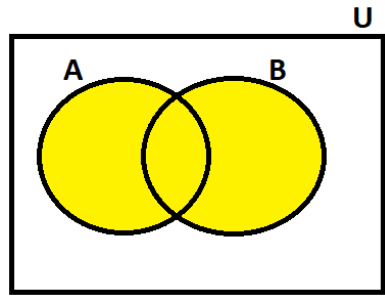
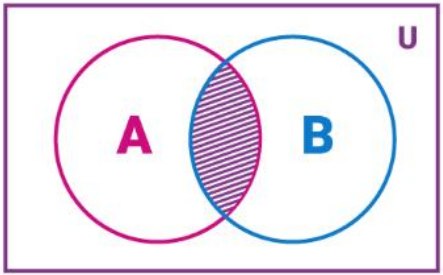
❑ **Assumption:** Image can be represented as a set of points.



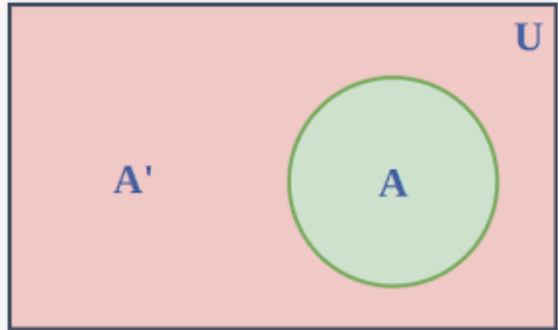
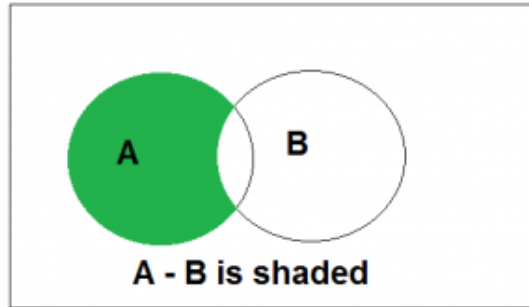
Once the image can be represented in the form of set of points, morphological operations on image are nothing but some set operations.

Set Theory Overview

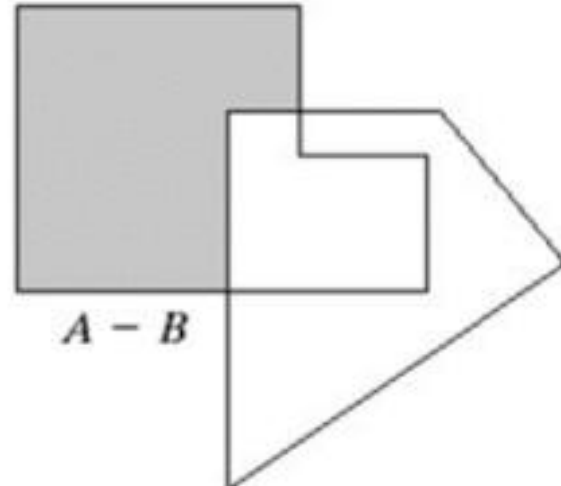
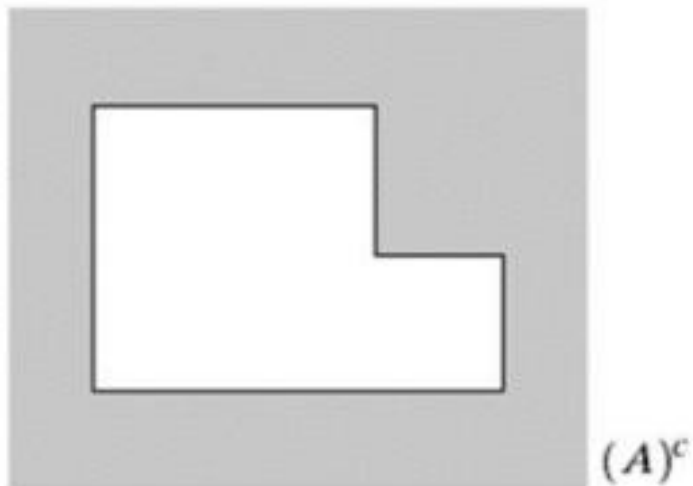
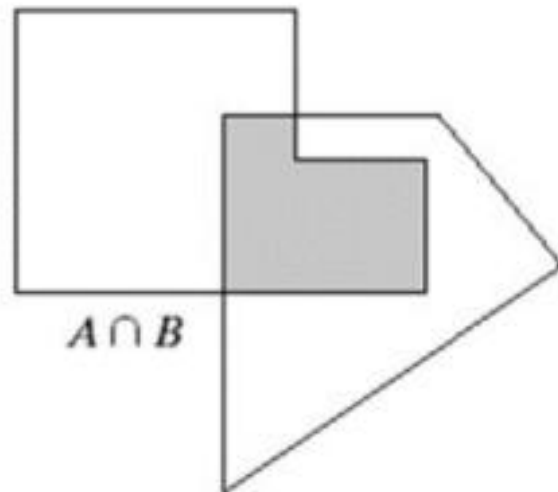
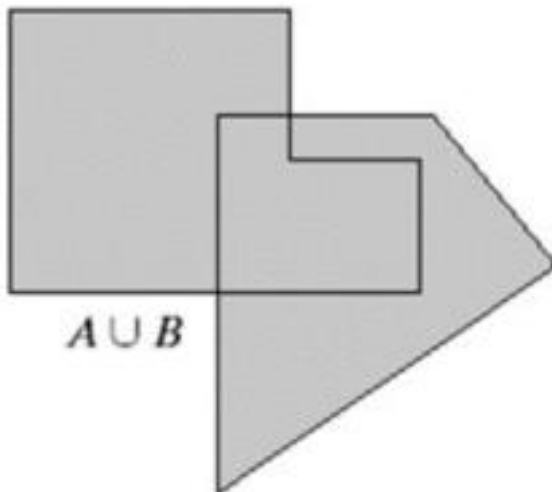
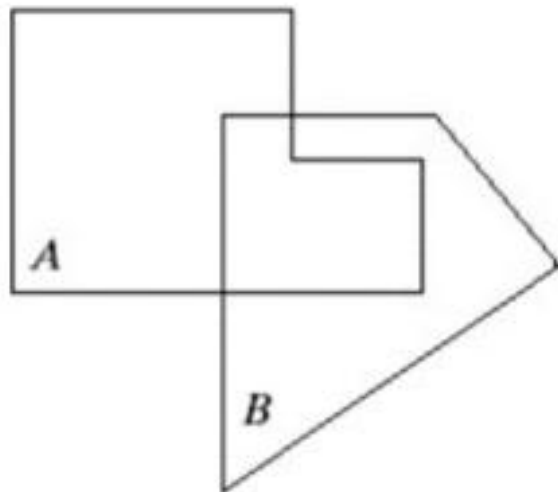
A in $\mathbb{Z}^2$ (2D), $a = \{a_1, a_2\}$, $a \in A$, if a is a member of point set A
 $b \notin A$, if b does not member of point set A .

Subset $A \subseteq B$	Every element of Set A must be a member of set B. Then set A is a subset of Set B.	
Union $A \cup B$	Set of elements taken from Set A and Set B	
Intersection $A \cap B$	Set of elements which are common in both Set A and Set B	

Morphological Image Processing

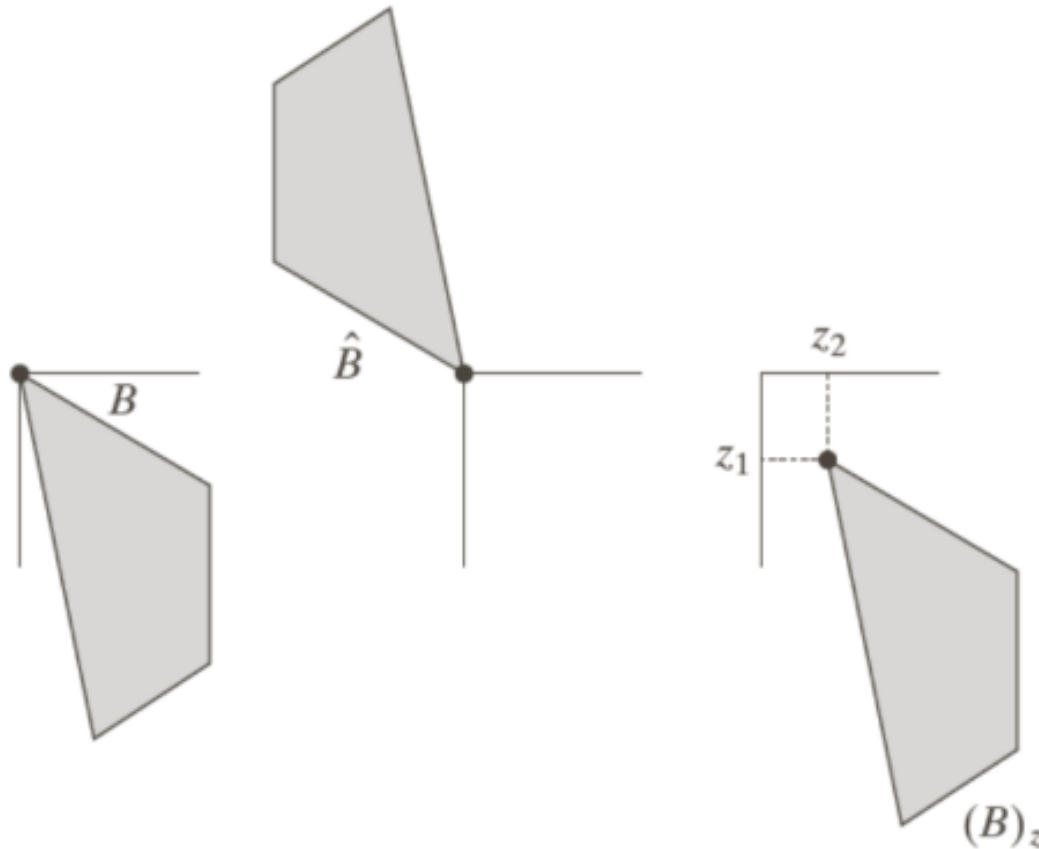
Complement A^C	Set of elements which does not belong to A $= \{b \mid b \notin A\}$	
Difference $A - B$	Set of element W such that W belongs to A and W does not belong to B $= \{w \mid w \in A \text{ and } w \notin B\}$	

Set Operation in Image Processing



Set Operation in Image Processing

- ❑ **Reflection** – The reflection of a set B , denoted by $\hat{B}$, is defined as $\hat{B} = \{w | w = -b, \text{ for } b \in B\}$
- ❑ **Translation** – The translation of a set B by point $z = (z_1, z_2)$, denoted by $(B)_z$ is defined as $(B)_z = \{c | c = b + z, \text{ for } b \in B\}$



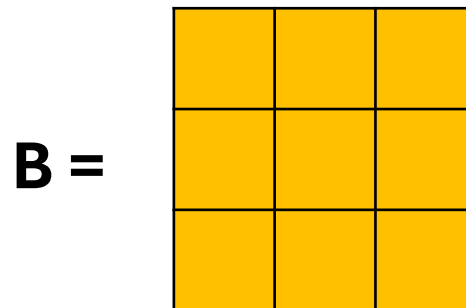
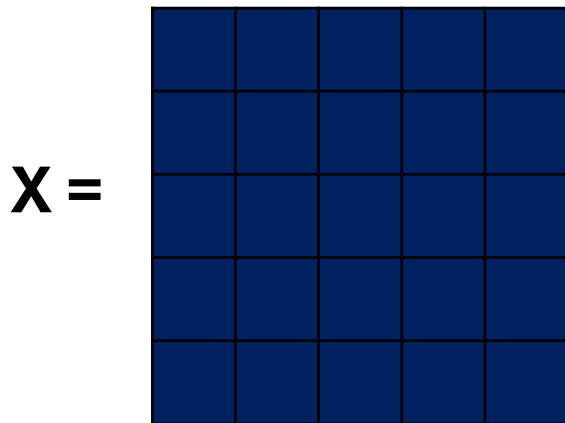
Morphological Transformation

❑ **Morphological Transformation:** Gives a relation of the image X , with another small point set B , called structuring element.

❑ In image processing, we use morphology with two types of sets of pixels (a) **Objects** and (b) **Structuring Elements**

○ Typically, **Objects** are defined as a sets of foreground pixels.

○ **Structuring elements (SE's)** be specified in terms of both foreground and background pixels.



Morphological Transformation

- ❑ The way we do this computation, is the similar way we have done the convolution operation.
- **In case of convolution,**
 - We have given a big image and a small image.
 - In convolution, we take this small image and move this small image systematically over the big image and for each shift or each such translation of the small image over this bigger image
 - which is the Convolution Operation.

Morphological Transformation

■ In morphological operation,

- we systematically move the structuring element B over the given image X
- and for each translation of the structuring element B in the given image we will perform some operation which the type of operation will depend upon what type of transformation or morphological transformation that we want to do.

Dilation Operation

Dilation Operation

□ Dilation:

- Symbol: $\oplus$
- **Assuming** X : be the input digital image and
 B : be the structuring element.

- First Definition:

$$X \oplus B = \{P \in Z^2 \mid P = x + b \text{ where } x \in X, b \in B\}$$

- Second Definition:

$$X \oplus B = \{P \mid (\hat{B})_p \cap X \neq \varphi\}$$

X

Dark Blue	Dark Blue	Dark Blue	Dark Blue	Dark Blue	Dark Blue
Dark Blue	Dark Blue	Dark Blue	Dark Blue	Dark Blue	Dark Blue
Dark Blue	Dark Blue	Dark Blue	Dark Blue	Dark Blue	Dark Blue
Dark Blue	Dark Blue	Dark Blue	Yellow	Dark Blue	Dark Blue
Dark Blue	Dark Blue	Dark Blue	Dark Blue	Dark Blue	Dark Blue
Dark Blue	Dark Blue	Dark Blue	Dark Blue	Dark Blue	Dark Blue

B

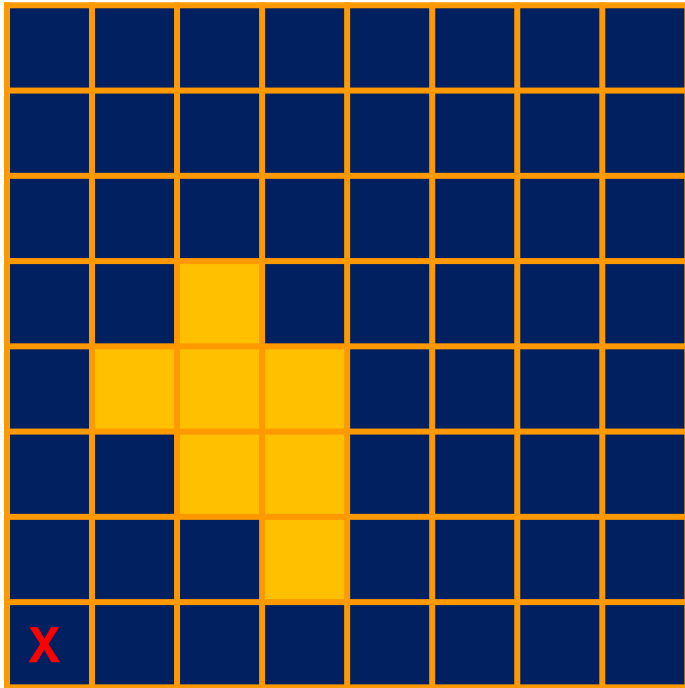
A 6x6 grid of squares. The central square, located at the intersection of the third and fourth columns and the third and fourth rows, is highlighted in yellow and contains a large red 'X'. All other squares in the grid are dark blue.
$$\mathbf{X} \oplus \mathbf{B}$$

The diagram illustrates the process of finding a path from a start cell to a goal cell on an 8x8 grid. Grid A shows a path of yellow cells from the bottom-left to the top-right. Grid B shows the same path with the start cell highlighted in red.

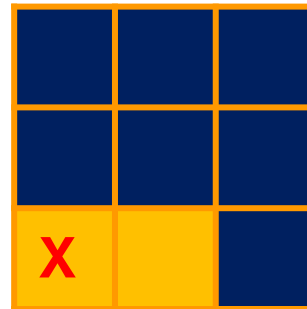
Dilation Operation

□ Example 2: Solution

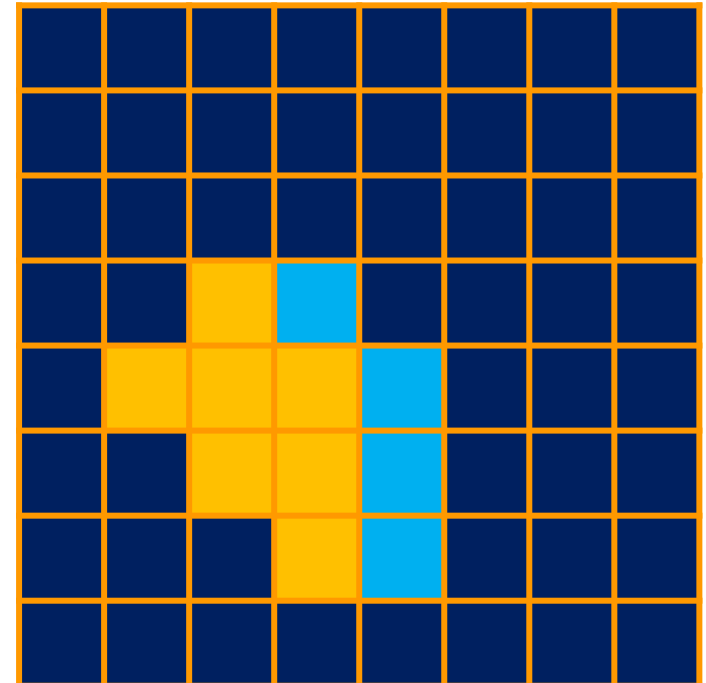
X



B

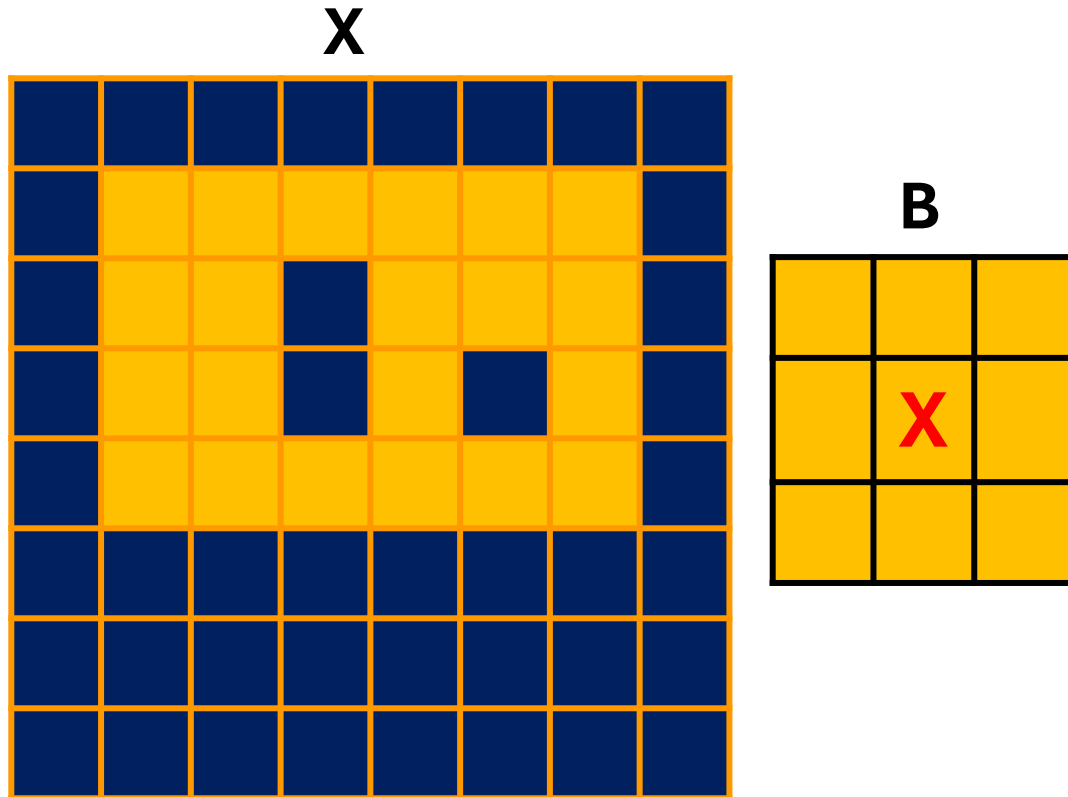


$X \oplus B$



Dilation Operation

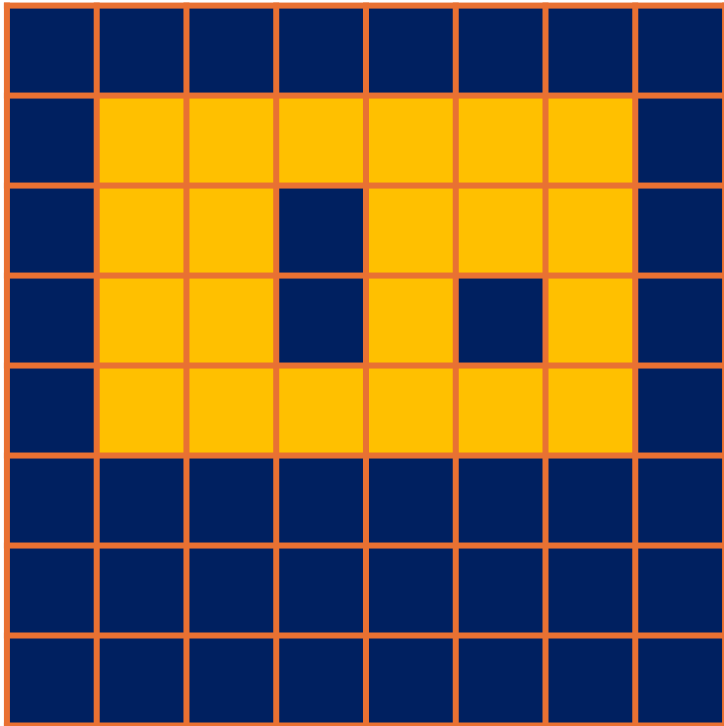
❑ **Example 3: Perform $X \oplus B$**



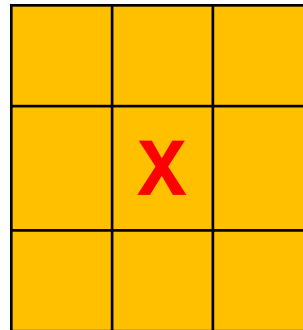
Dilation Operation

□ Example 3: Solution

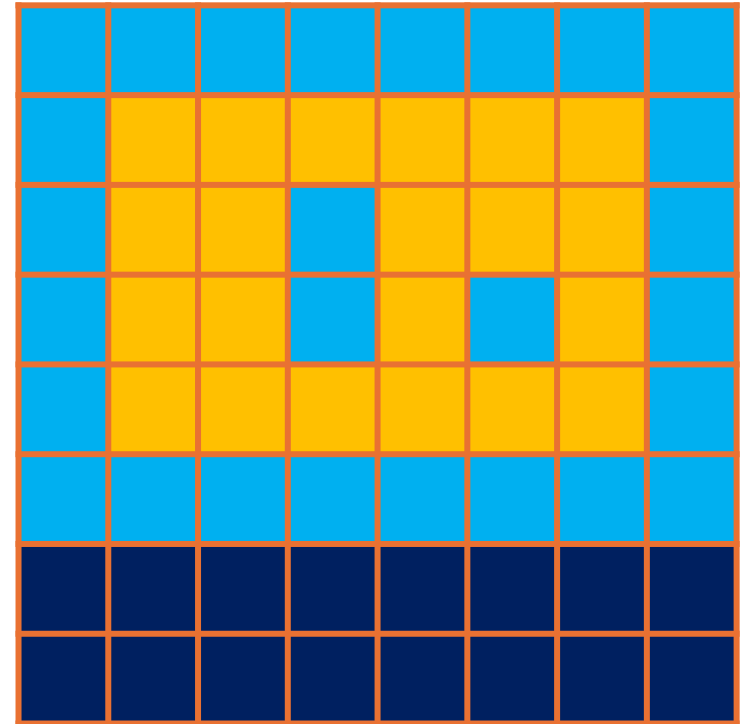
X



B



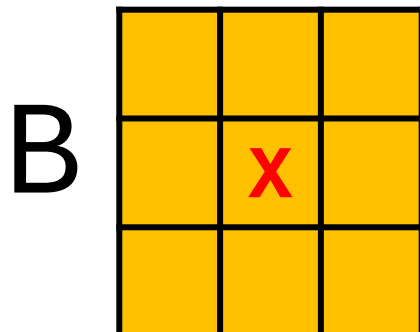
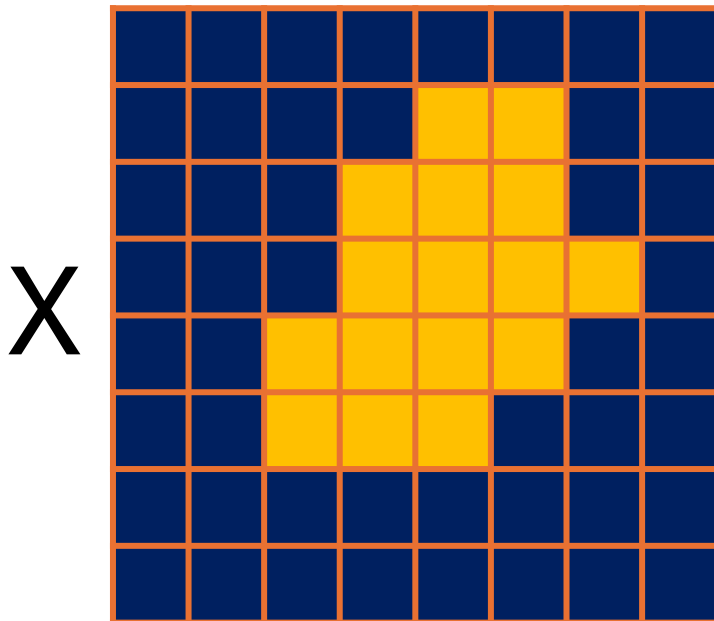
$X \oplus B$



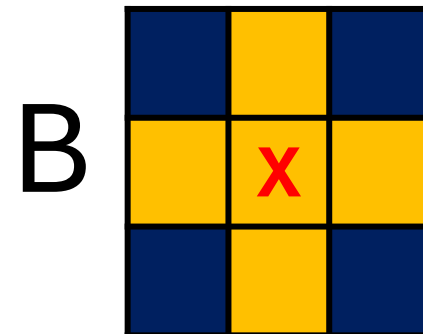
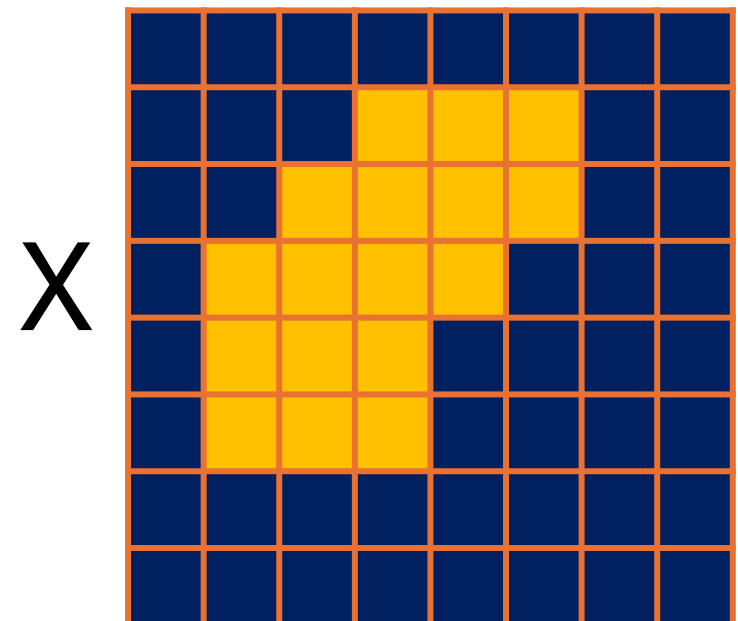
Dilation Quiz

Perform $X \oplus B$

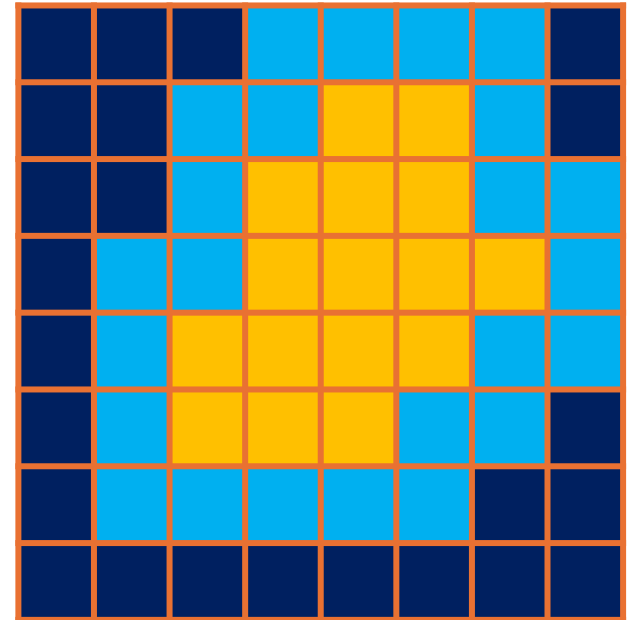
Ques1



Ques 2



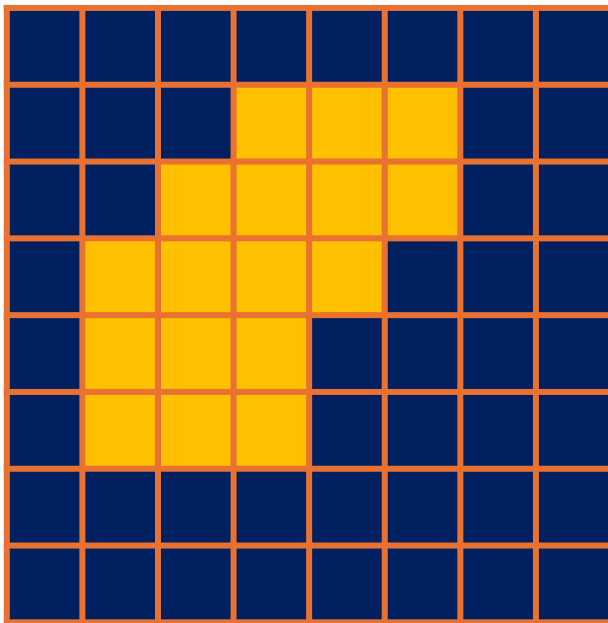
X



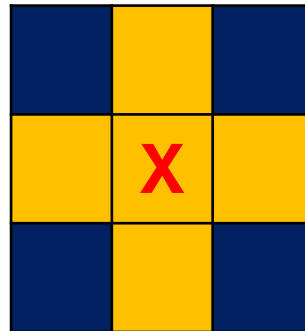
Dilation Quiz (Solution)

Ques 2 Perform $X \oplus B$

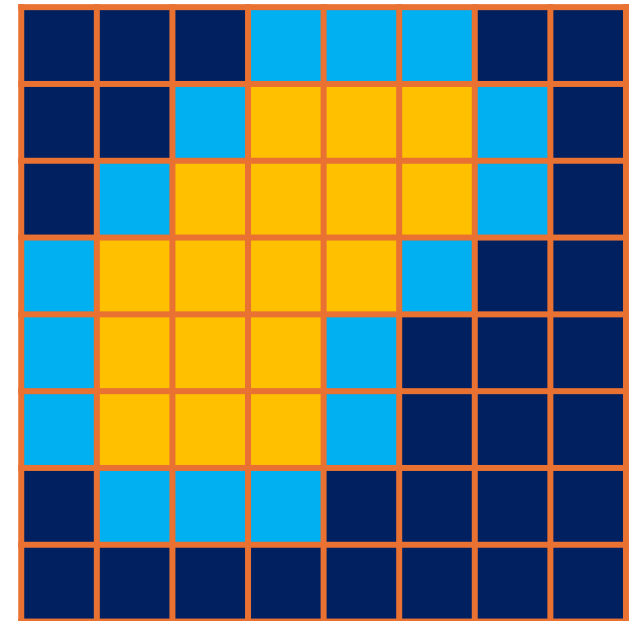
X



B



$X \oplus B$



Dilation Operation

□ Dilation:

- Dilation is an operation that grows or thickens objects in a binary image
- The specific manner of this thickening is controlled by a shape referred to as a structuring element
- The structuring element is translated throughout the domain of the image to see where it overlaps with 1-valued pixels
- The output image is 1 at each location of the origin such that the structuring element overlaps at least one 1-valued pixel in the input image

Dilation Operation

❑ Application of Dilation Operation:

- To fill the black spot, ideally which should be the white spot. This operation dilation can be used.
- Dilation can repair breaks.



- Dilation can repair intrusions.



- However, the side-effect is ***“Boundary of original image is expanded”***.

Properties of Dilation Operation

- Dilation is commutative

$$\mathbf{A \oplus B = B \oplus A}$$

- Dilation is associative

$$\mathbf{A \oplus (B \oplus C) = (A \oplus B) \oplus C}$$

- Dilation is invariant to translation

$$\mathbf{A_h \oplus B = (A \oplus B)_h}$$



Erosion Operation

Erosion Operation

□ Symbol: $\ominus$

□ First Definition:

$$X \ominus B = \{p \in \mathbb{Z}^2 \mid p + b \in X \text{ for every } b \in B\}$$

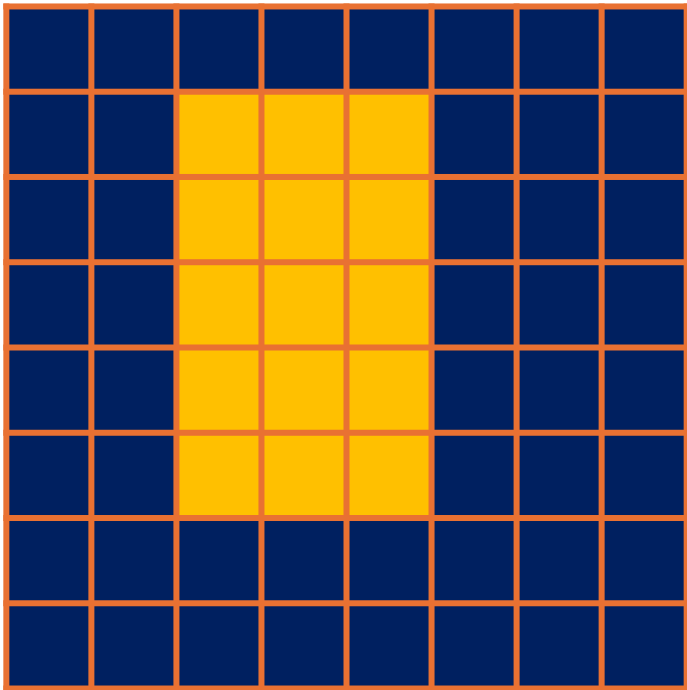
□ Second Definition:

$$X \ominus B = \{p \mid (B)_p \subseteq X\}$$

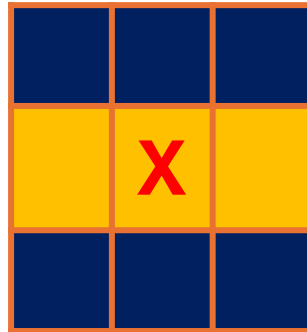
Erosion Operation

❑ Example 1: Perform $X \ominus B$

X



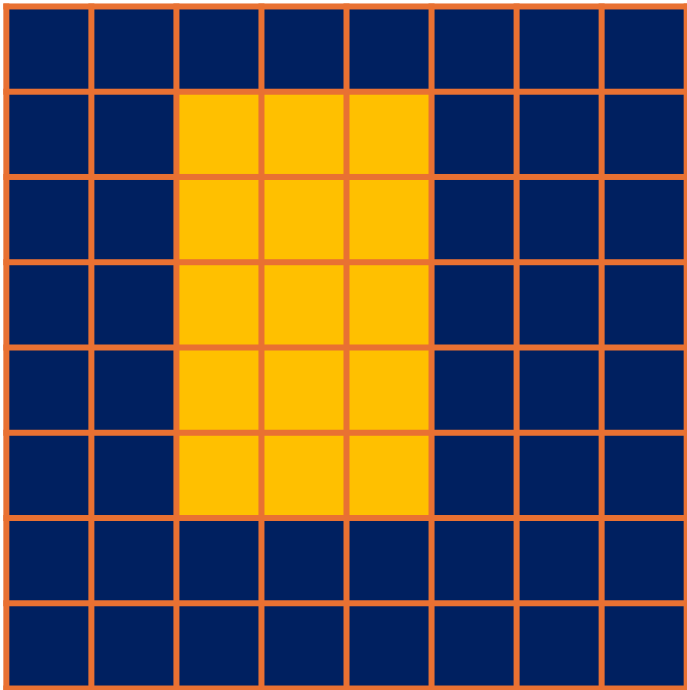
B



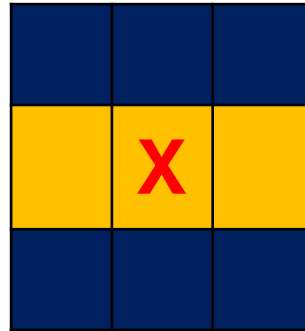
Erosion Operation

❑ Example 1: Perform $X \ominus B$

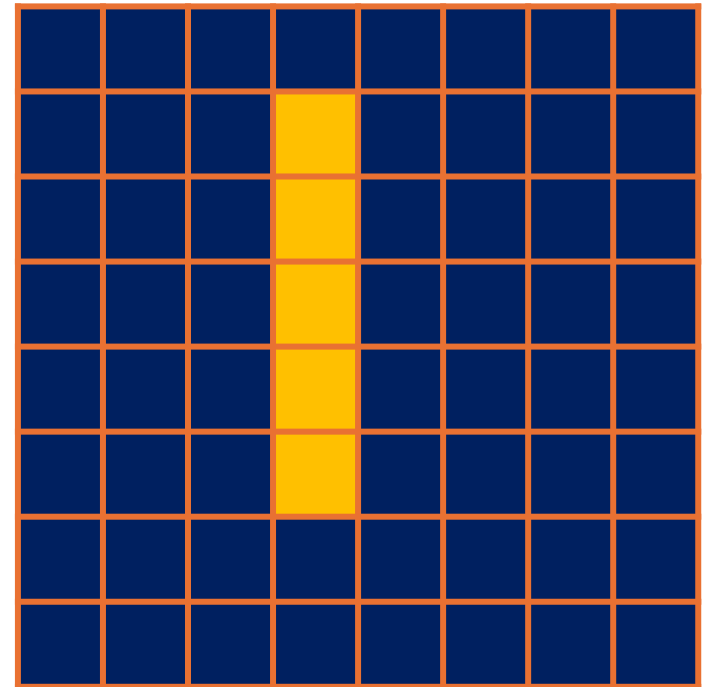
X



B



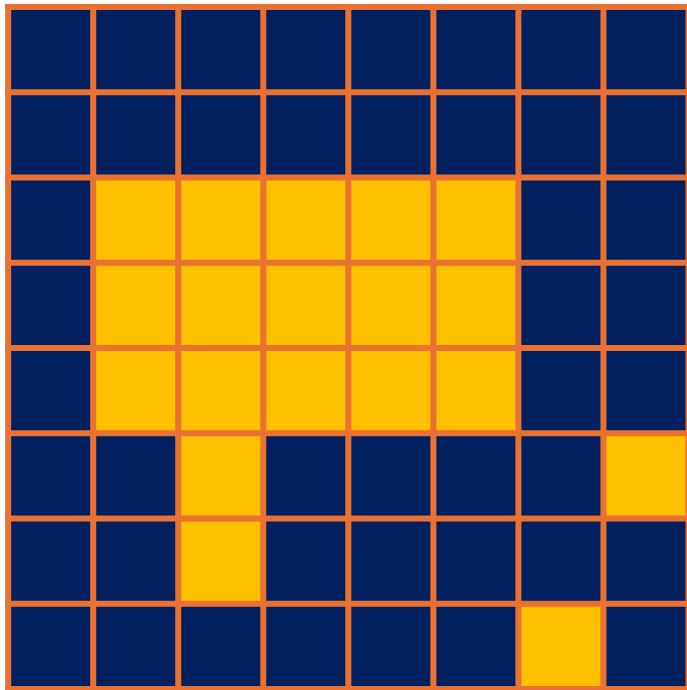
$X \ominus B$



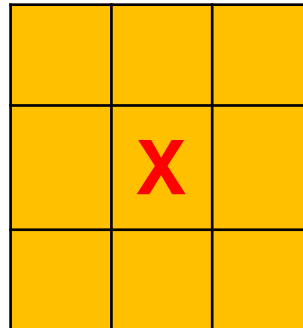
Erosion Operation

❑ Example 2: Perform $X \ominus B$

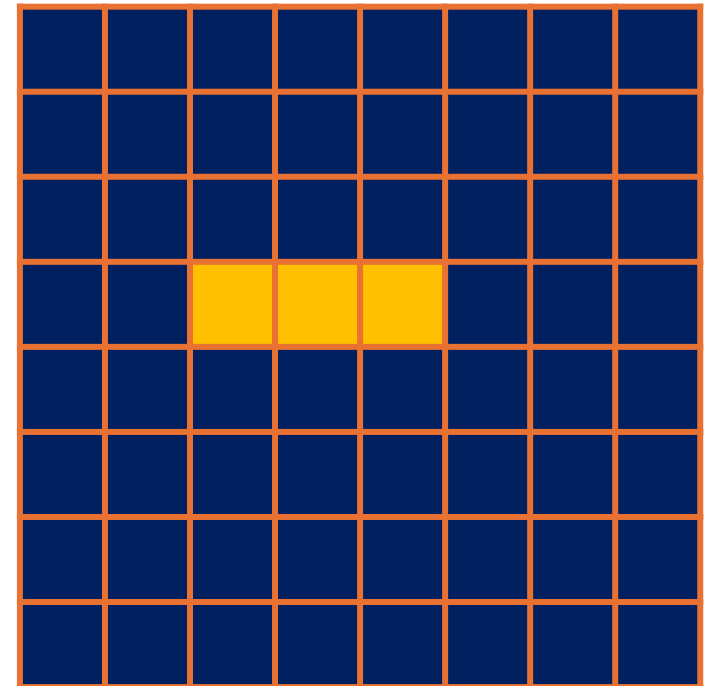
X



B



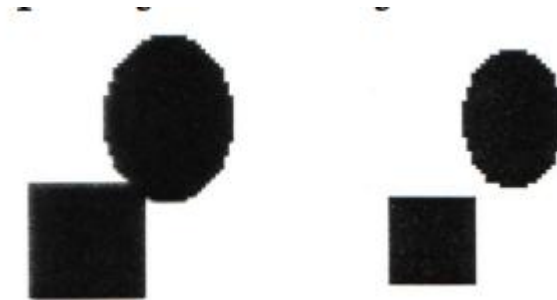
X



Erosion Operation

□ Application of Erosion Operation:

- Erosion can split apart joined objects.



- Erosion shrinks objects and removes random outer edges.

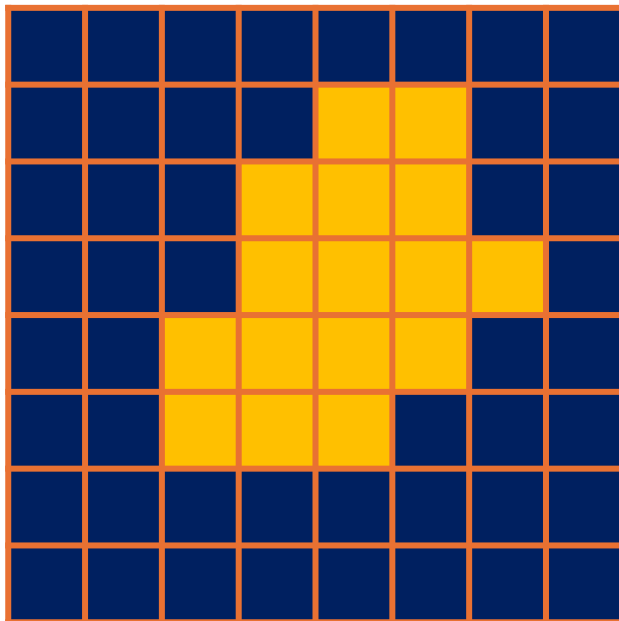


Erosion Quiz

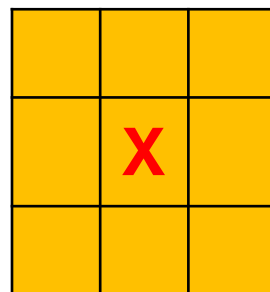
Perform $X \ominus B$

Ques1

X

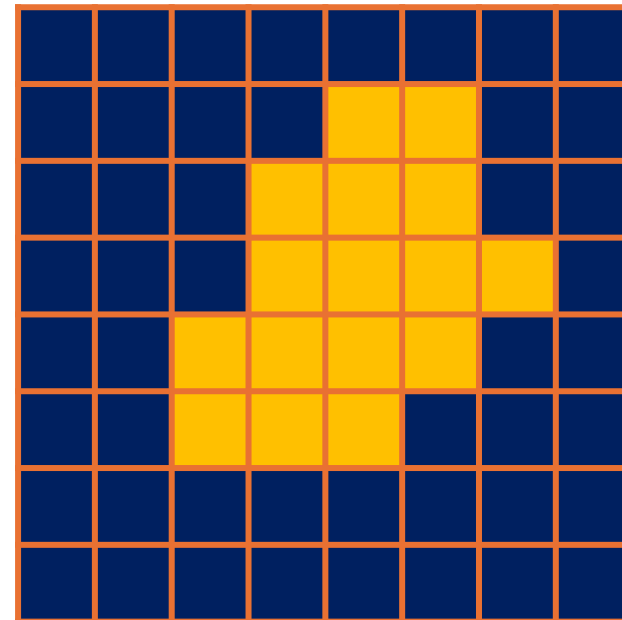


B

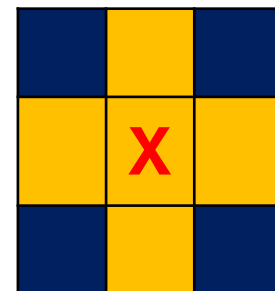


Ques2

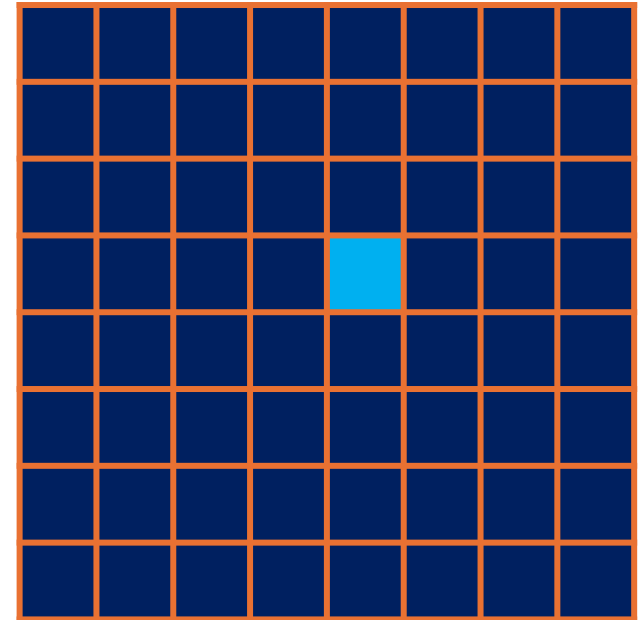
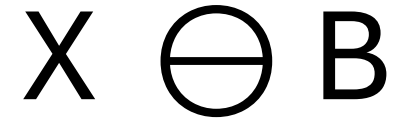
X



B



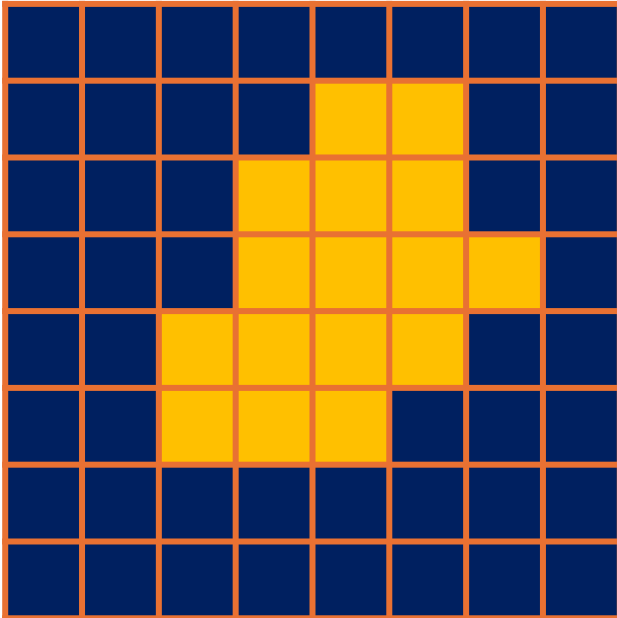
X



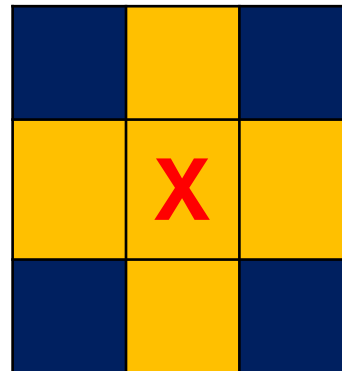
Erosion Quiz (Solution)

Ques2: Perform $X \ominus B$

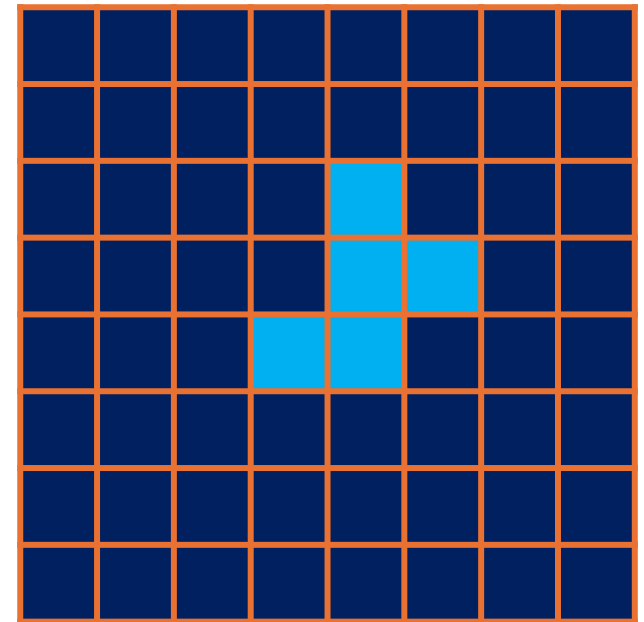
X



B



$X \ominus B$



Duality of Dilation and Erosion

- Erosion and dilation are duals of each other with respect to set complementation and reflection:

$$(A \ominus B)^c = \{z | (B)_z \subseteq A\}^c$$

$$= \{z | (B)_z \cap A^c = \emptyset\}^c$$

$$= \{z | (B)_z \cap A^c \neq \emptyset\}$$

$$= A^c \oplus B$$

Summary of Dilation and Erosion

❑ **In Dilation:** object region gets expanded

And **In Erosion:** object region gets reduced. So, these are opposite operations.

❑ So, if I apply dilation operation then the resultant image must be compensated by erosion operation and vice-versa.

▪ **Opening Operation:** $A \circ B = (A \ominus B) \oplus B$

▪ **Closing Operation:** $A \cdot B = (A \oplus B) \ominus B$

Combining Dilation and Erosion Operation

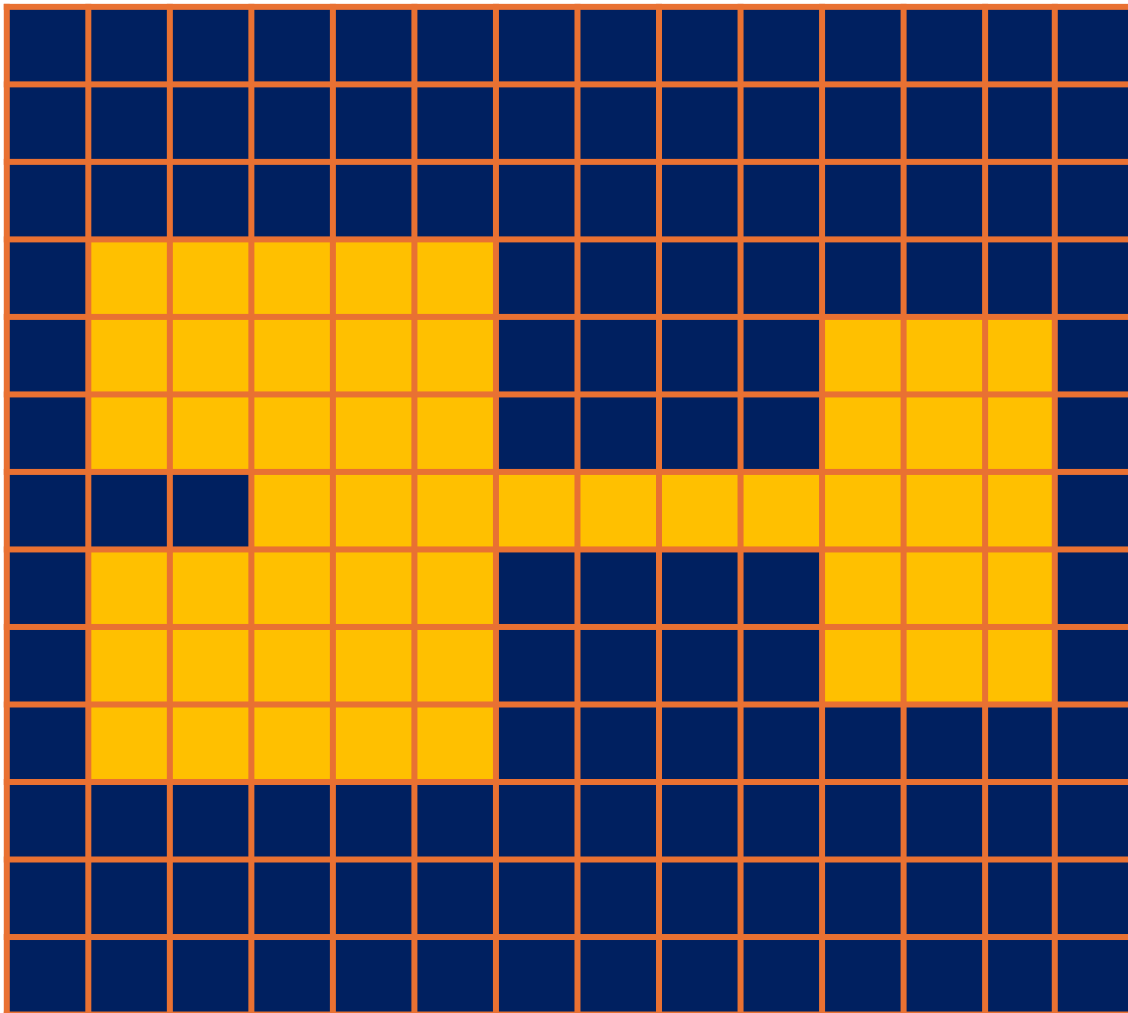
- Dilation and Erosion are not inverse transformations
- If an image is eroded & then dilated (or vice-versa), the original image can not be obtained
- In practical applications, dilation and erosion are used most often in various combinations
- Three of the most common combinations of dilation and erosion are
 - Opening
 - Closing
 - Hit or miss transformation

Opening Operation

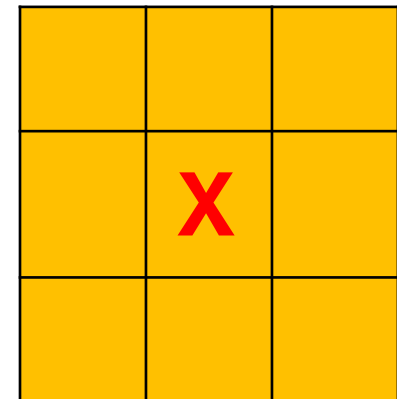
Opening Operation

❖ Opening Operation: $A \circ B = (A \ominus B) \oplus B$

A

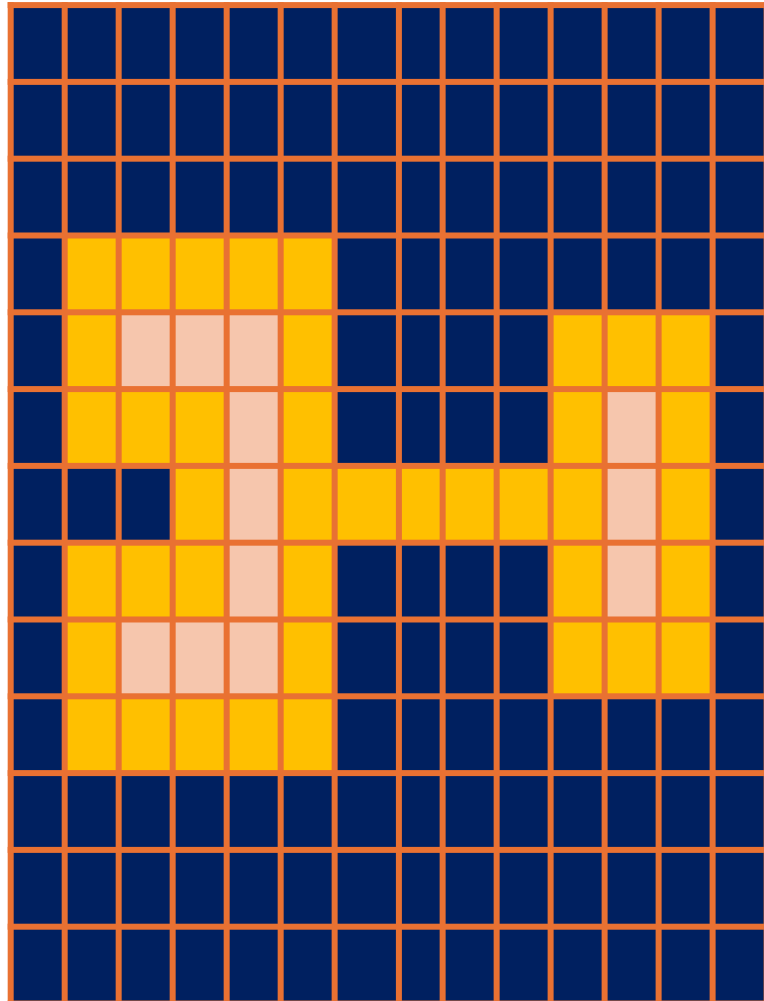


B

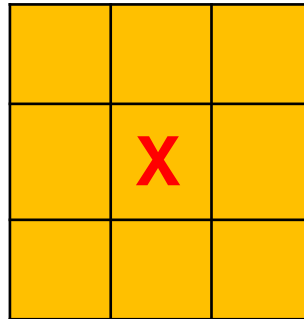


Opening Operation

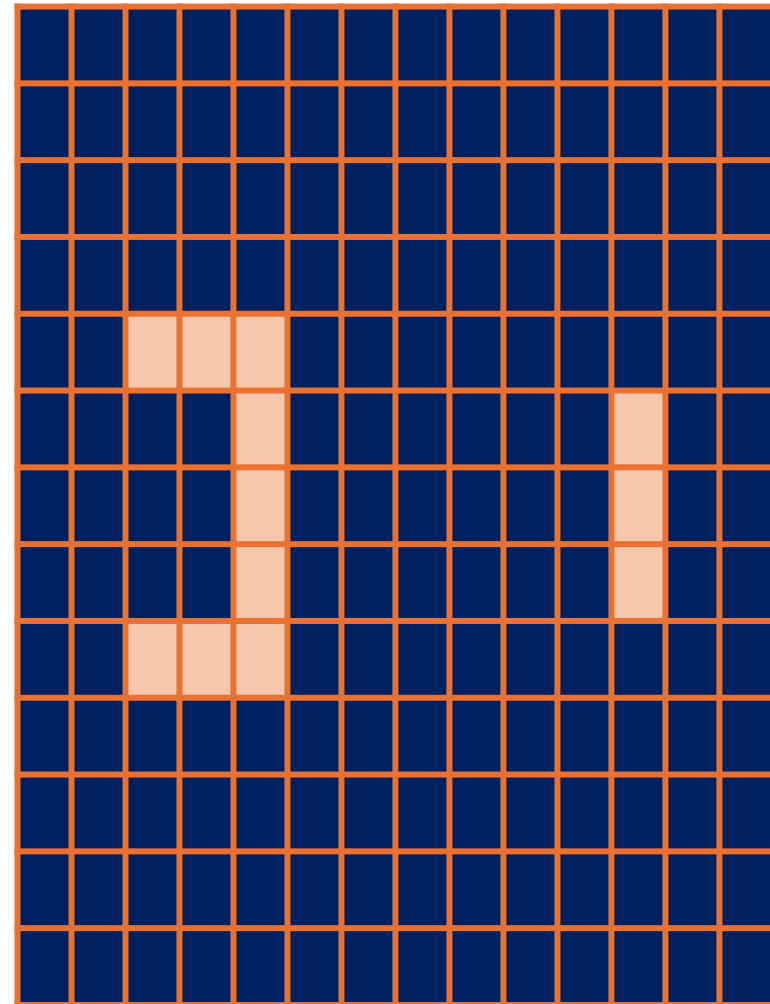
A



B

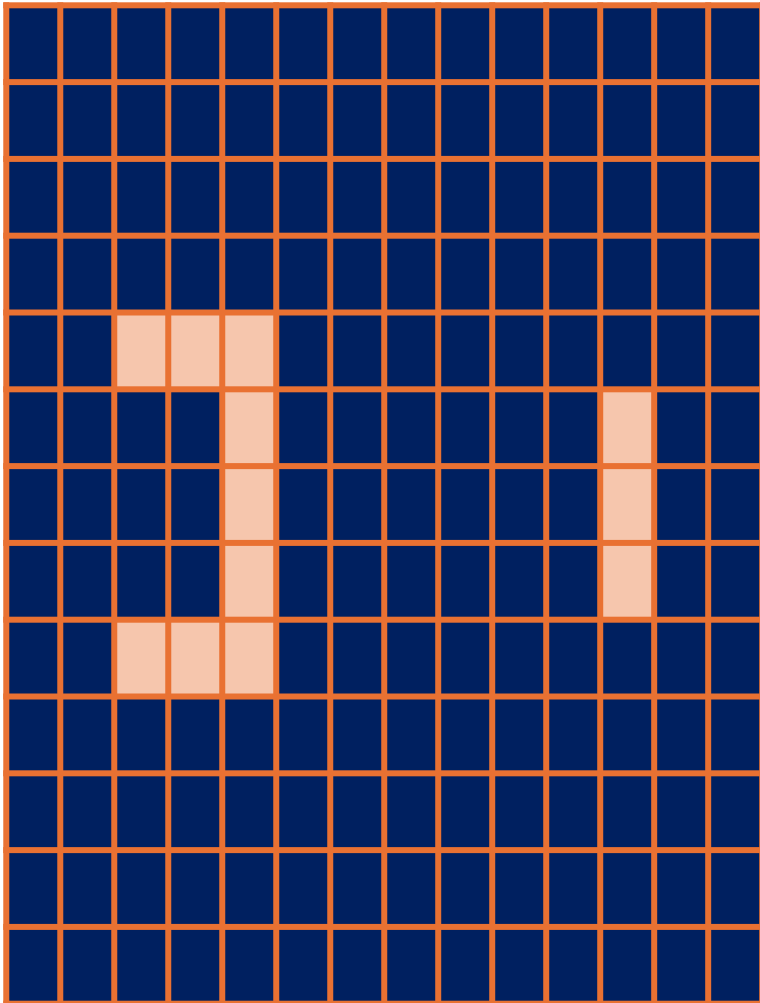


$(A \ominus B)$

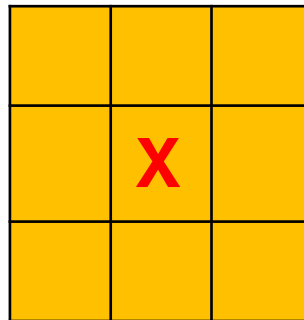


Opening Operation

$$(A \ominus B)$$

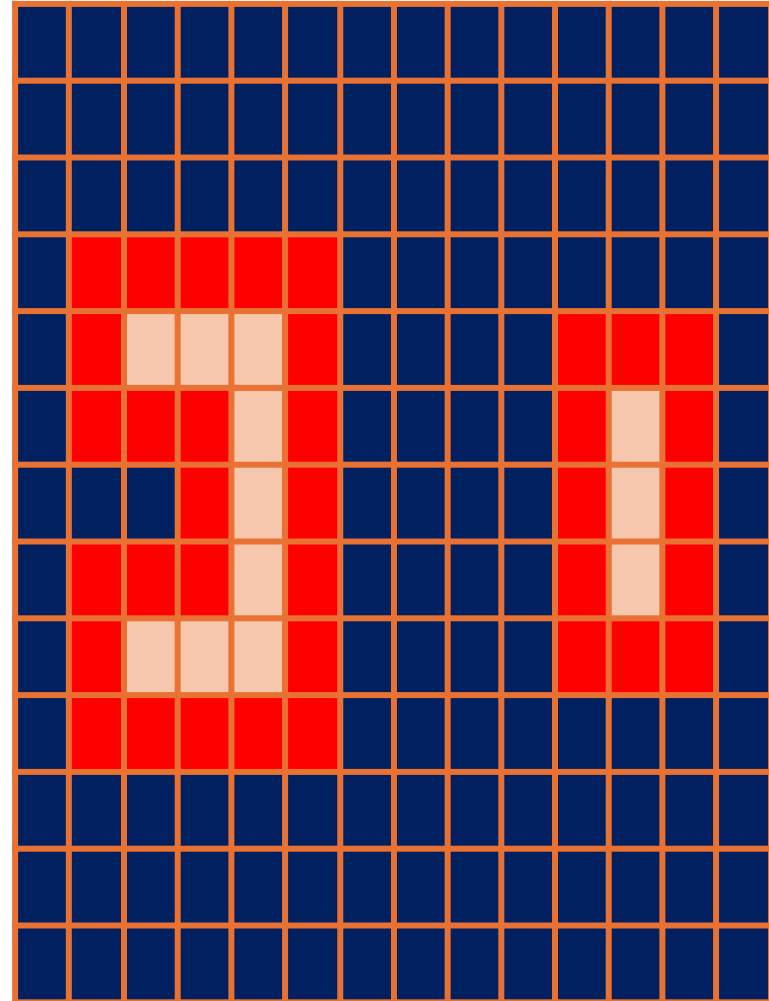


B



Opening

$$A \circ B = (A \ominus B) \oplus B$$



Opening Operation

❖ The objectives of Opening operation are

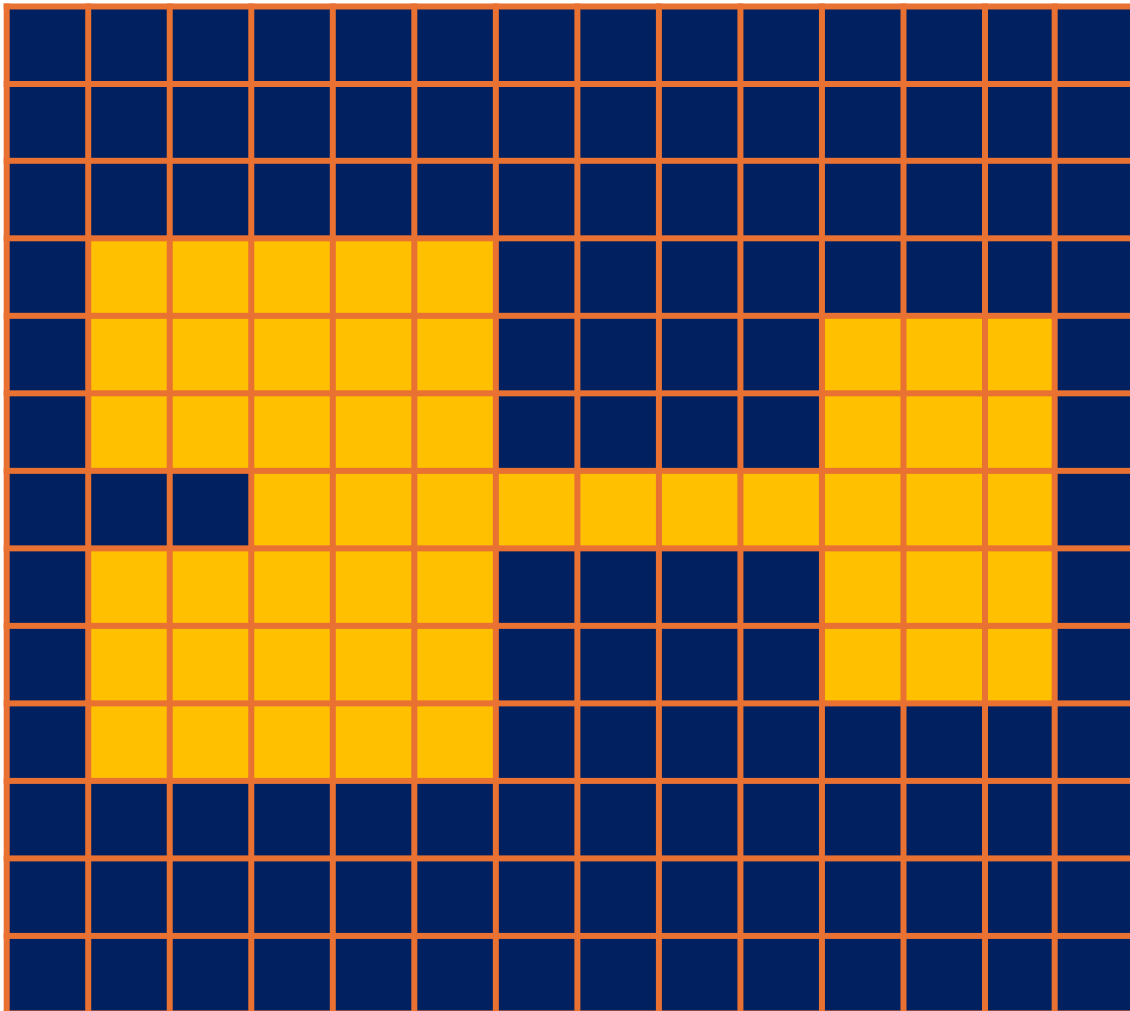
- Smooth the contours of an object
- Breaks narrow strips
- Eliminates thin edges
- It is less destructive than the Erosion

Closing Operation

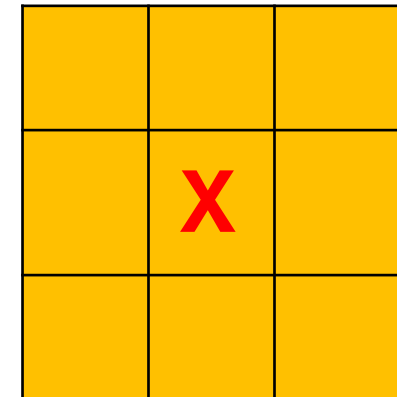
Closing Operation

❖ Closing Operation: $A \cdot B = (A \oplus B) \ominus B$

A

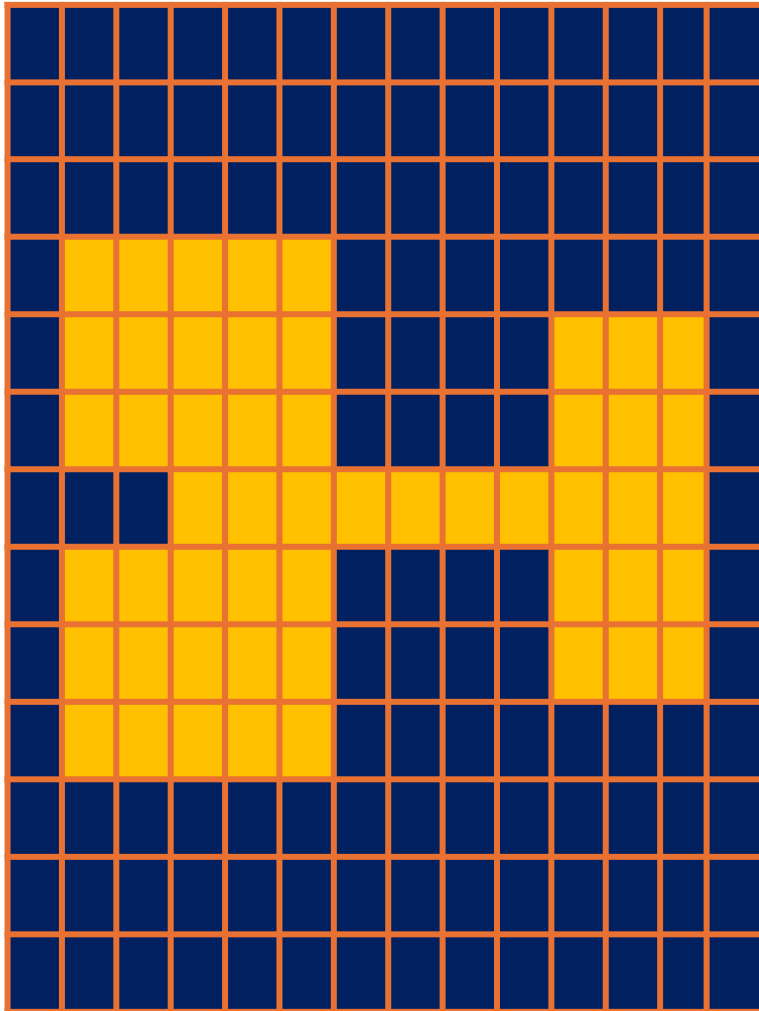


B

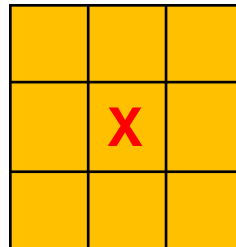


Closing Operation

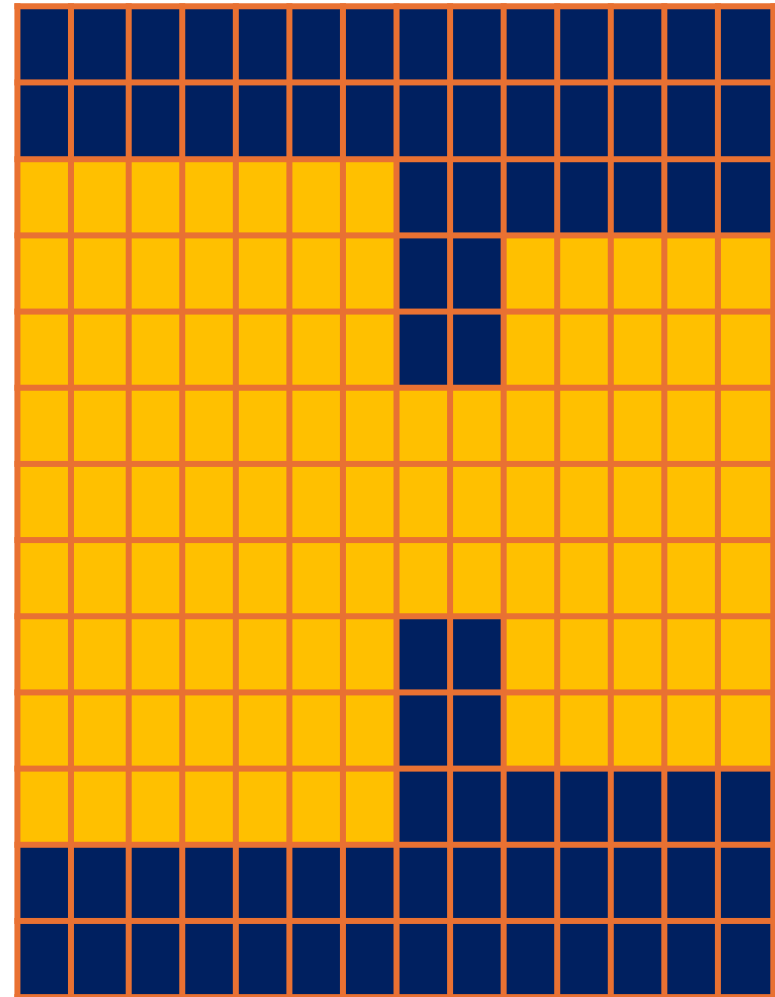
A



B

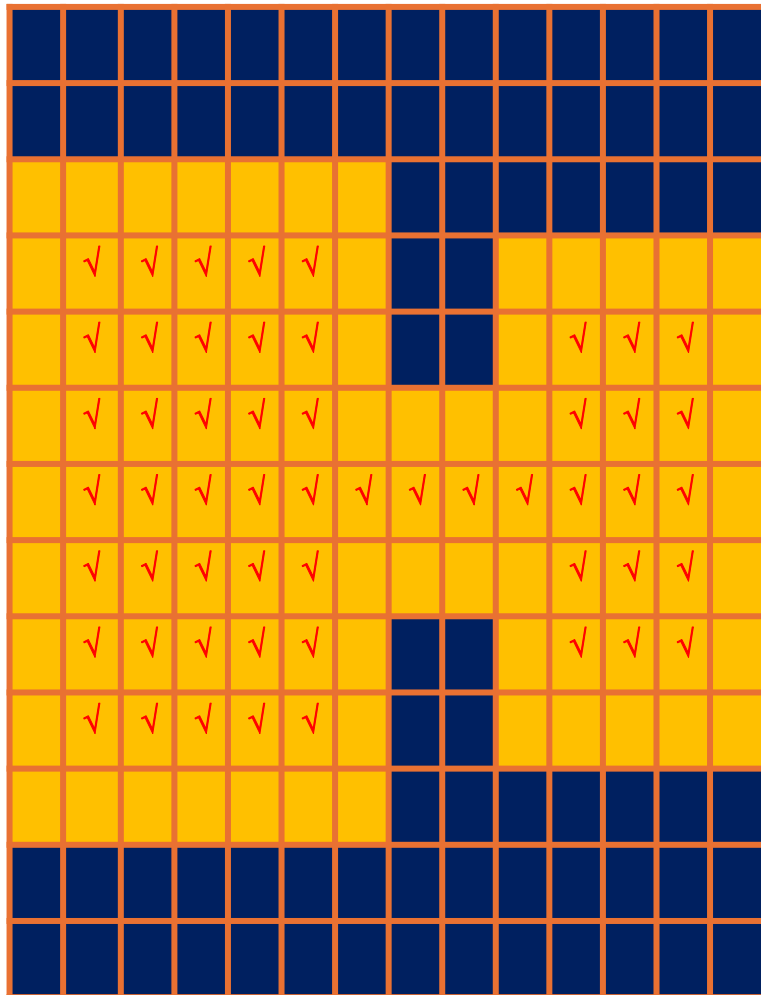


$(A \oplus B)$

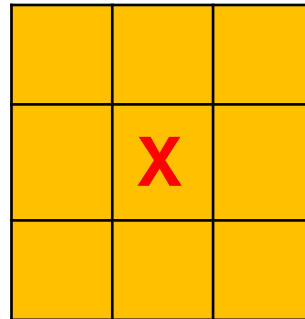


Closing Operation

$$(A \oplus B)$$

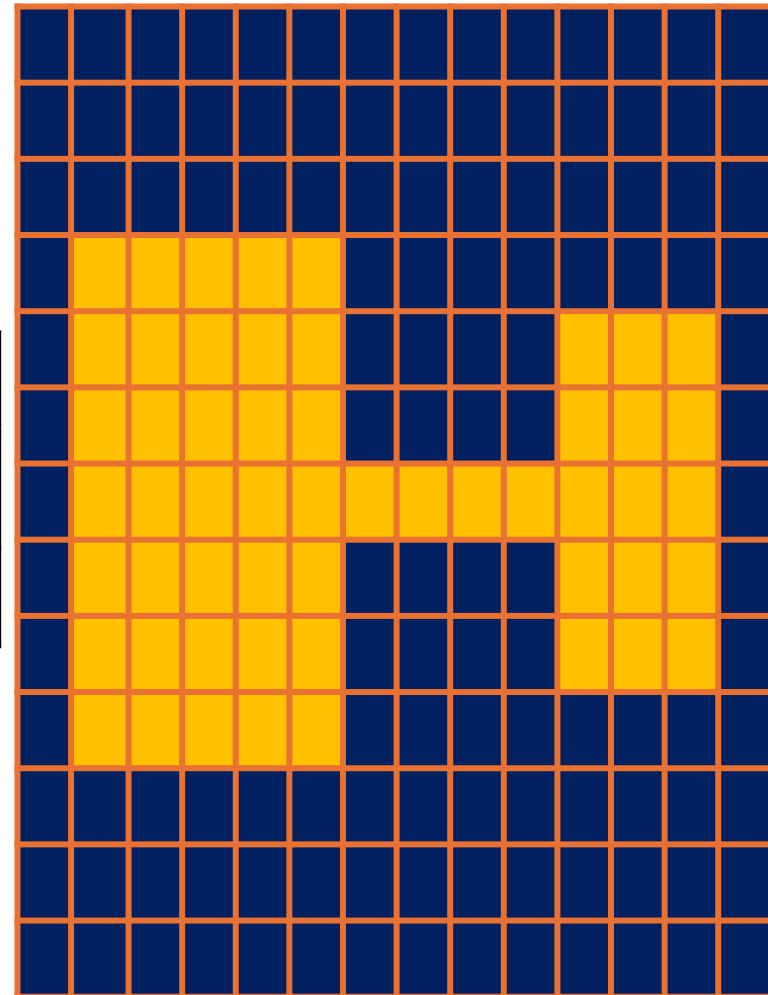


B



Closing

$$A \cdot B = (A \oplus B) \ominus B$$



Previous Year Questions

Question on Dilation and Erosion

Ques1: An image of size 5 x 4 and structuring element of size 1 x 2 are given below. Apply erosion, dilation operation to find the result.

GLA University Mathura End Term Examination, Even Semester, 2023-24

GLA University Mathura End Term Examination, Odd Semester, 2023-24

GLA University Mathura, Summer Carry Examination, Session 2023-24

1			
1			
	1	1	1
	1		
	1		

<u>1</u>	1
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Question on Dilation and Erosion

Ques 2: Perform Erosion and Dilation on the image shown in Figure. Use 0 value for padding.

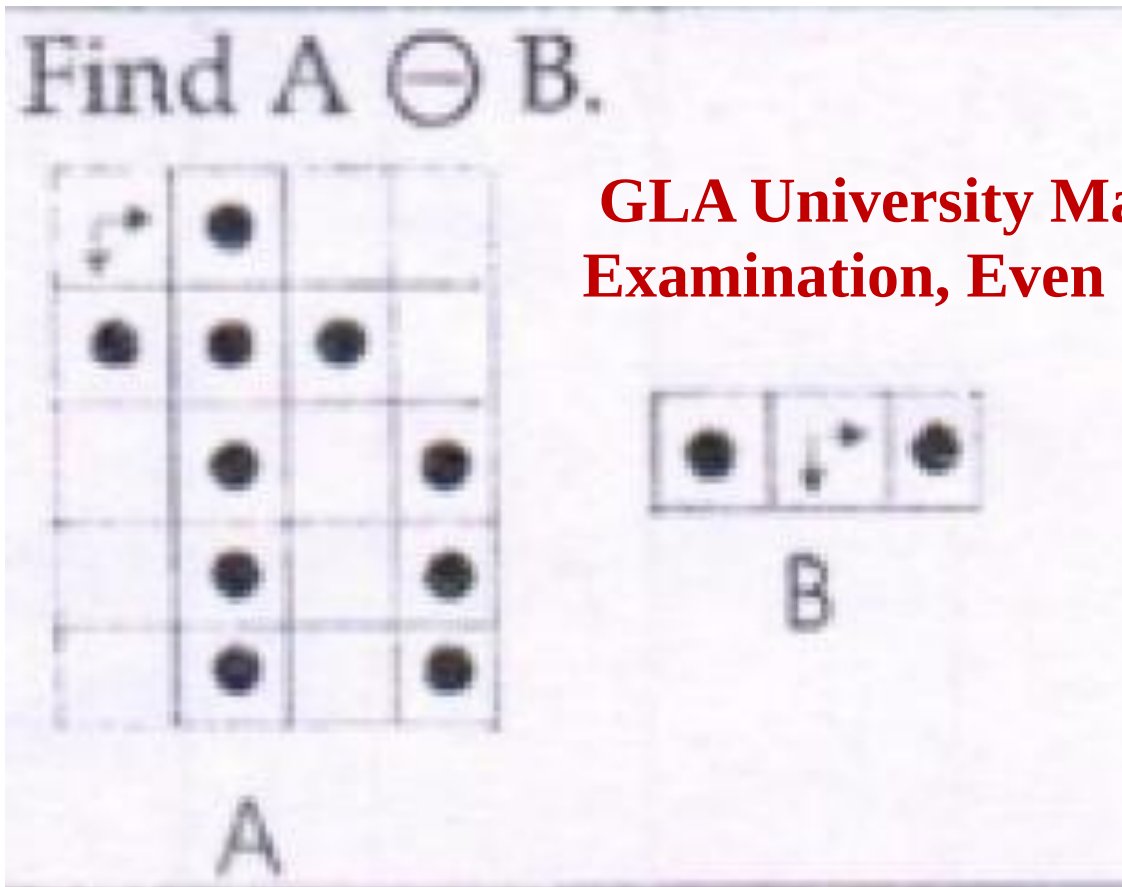
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0	0	0	0	0
0	0	1	0	0
0	1	1	1	0
0	1	0	0	0
0	0	0	0	0

0	0	0
1	1	1
0	1	0

Question on Dilation and Erosion

Ques 3:



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Question on Dilation and Erosion

Ques 4:

Binary image X and structuring element B, are given in Figure.1:

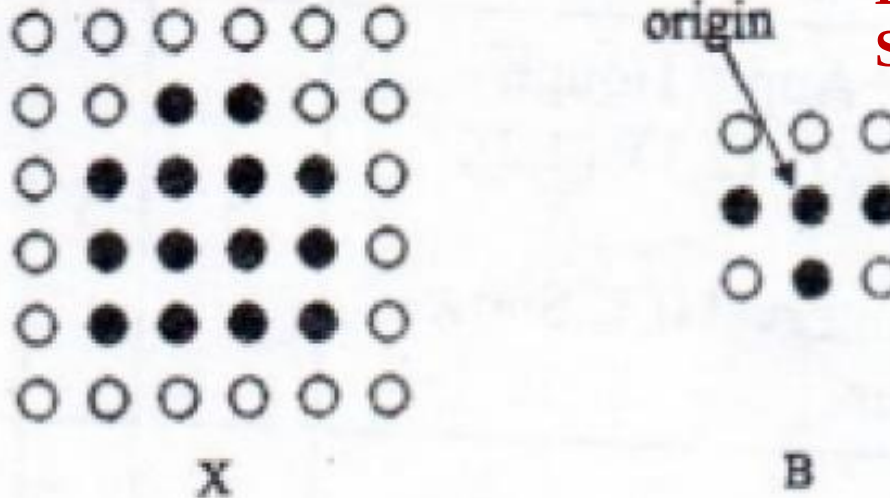


Figure:1

Calculate $Y1 = X \ominus B$, where $\ominus$ denotes the morphological erosion operator and $Y2 = X \oplus B$ where $\oplus$ denotes the morphological dilation operator;

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Questions on Opening and Closing

Ques 5:

b. Apply the opening operation on the image shown on the left using the structuring element on the right. Show all the steps in 8×8 grids.

		1	1	1	1	1	
		1	1	1			
	1	1	1				
		1	1				

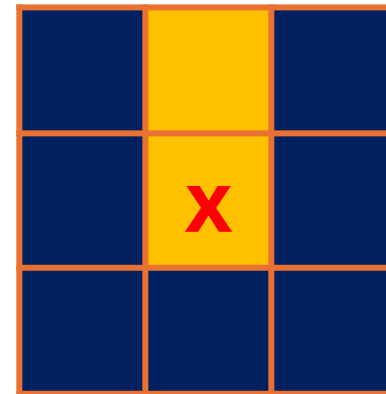
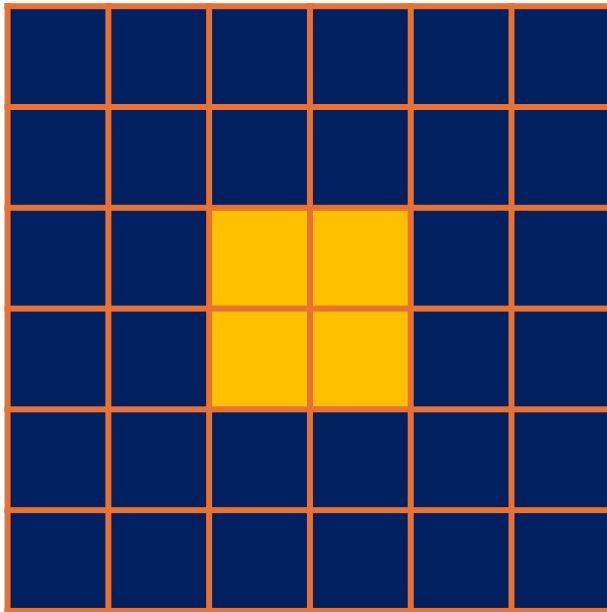
	1	
1	1	1
	1	

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Question on Dilation and Erosion

Ques 6: Given a binary image I_0 as shown in Fig 6(a) and a structuring element S as shown in Fig 6(b) in which the shaded area represent the object pixel and x is the origin.

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Find the diluted image using the formula

$$I \oplus S = \{p | p = i + s, i \in I, s \in S\}$$

Question on Dilation and Erosion

Ques 2: Perform Erosion and Dilation on the image shown in Figure. Use 0 value for padding.

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0	0	0	0	0
0	0	1	0	0
0	1	1	1	0
0	1	0	0	0
0	0	0	0	0

1	<u>1</u>	1
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Thank You